



# Statistics of Financial Markets

## Chapter 9: ARCH and GARCH models



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## Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AR**: AutoRegressive (model); Abnormal Return
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARCH-LM**: ARCH Lagrange Multiplier (test)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **EGARCH**: Exponential GARCH
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **ES**: Expected Shortfall
- ▣ **ETF**: Exchange-Traded Fund
- ▣ **EWMA**: Exponentially Weighted Moving Average
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

## Acronyms used in this material (2/2)

- ▣ **GJR**: Glostén–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **GJR-GARCH**: Glostén–Jagannathan–Runkle GARCH
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **IGARCH**: Integrated GARCH
- ▣ **LB**: Ljung–Box (test)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MLE**: Maximum Likelihood Estimation
- ▣ **NIC**: News Impact Curve
- ▣ **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- ▣ **QLIKE**: Quasi-LIKelihood (loss function)
- ▣ **QMLE**: Quasi-Maximum Likelihood Estimation
- ▣ **QQ**: Quantile–Quantile (plot)
- ▣ **RW1**: Random Walk 1 (i.i.d. increments)
- ▣ **RW3**: Random Walk 3 (uncorrelated increments)
- ▣ **SE**: Standard Error
- ▣ **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

## Today's Question and Route

- **Question:** if tomorrow's return cannot be predicted, can tomorrow's **risk** be predicted?
  - ▶ **Chapter 7:** daily returns are almost uncorrelated; **Chapter 8:** their squares are strongly correlated (volatility clustering)
  - ▶ this chapter turns volatility clustering into a model for the conditional variance
- **Route** of the chapter
  - ▶ conditional mean and conditional variance; ARCH( $q$ ) and GARCH(1,1)
  - ▶ estimation by maximum likelihood; Student-t and skewed-t innovations
  - ▶ six markets; asymmetry: GJR-GARCH, EGARCH, the news impact curve
  - ▶ diagnostics, model choice, forecasts, VaR 1%

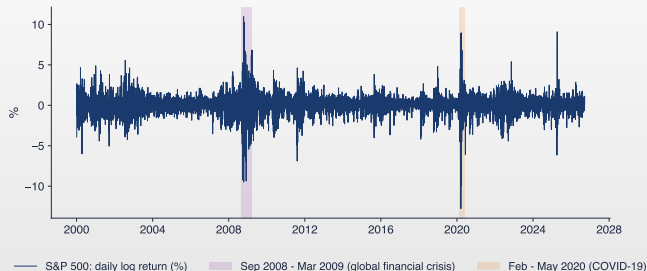
## Learning Outcomes

- ▣ Distinguish the conditional variance from the unconditional variance and explain why returns can be uncorrelated yet dependent
- ▣ Write the ARCH( $q$ ) and GARCH(1,1) models and compute the persistence, the unconditional variance and the half-life
- ▣ Estimate GARCH models by maximum likelihood, read the standard errors and choose the innovation distribution
- ▣ Measure the leverage effect with GJR-GARCH, EGARCH and the news impact curve
- ▣ Check a fitted model with standardised residuals and choose among models with AIC and BIC
- ▣ Forecast volatility over several horizons, compare forecasts with QLIKE and compute a one-day VaR 1%

## Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 13 (13.1 ARCH and GARCH models, 13.2 extensions of the GARCH model)
  - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 13
- Conditional heteroskedastic models: [Tsay \(2010\)](#), Ch. 3
- Python Quantlets of this chapter: [Quantlets/Ch\\_09](#)
  - ▶ ported from the SFE Quantlets `SFEtimegarch`, `SFElikarch1`, `SFElikgarch`, `SFEgarchest`, `SFEvolgarchest` and `SFENewsImpactCurve`
  - ▶ estimation with the Python package `arch`
- Lecture notebook: [open in Google Colab](#)
- Video course: [!\[\]\(5ba1bc70d78f05c00988641e5e513c62\_img.jpg\) \[Applied Time Series Analysis with Python\]\(#\)](#)

## S&P 500: Calm Years and Storms



- Daily log returns  $r_t = 100(\ln P_t - \ln P_{t-1})$ , in %, 6 714 days to 18 September 2026; the largest fall:  $-12.8\%$  on 16 March 2020; the largest rise:  $11.0\%$  on 13 October 2008
- Standard deviation:  $0.42\%$  in 2017,  $3.52\%$  from September 2008 to March 2009,  $3.70\%$  from February to May 2020
- Large changes follow large changes, of either sign: volatility clustering [Engle \(1982\)](#)

## Two Storms: 2008 and 2020



Lehman Brothers headquarters, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons



The Financial District of New York, 25 March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

- 15 September 2008: Lehman Brothers files for bankruptcy; March 2020: the COVID-19 pandemic closes economies
- In both episodes the daily standard deviation of the S&P 500 was 8.3 and 8.8 times that of 2017; calm returned only after months

## Why Model the Conditional Variance?

- ▣ **Risk management:** VaR (value at risk) and ES (expected shortfall) need tomorrow's volatility, not the average volatility of the last 20 years
  - ▶ a constant-variance VaR is too large in calm years and too small in crises ([Chapter 10](#))
- ▣ **Portfolios:** weights depend on variances and covariances, which change over time
- ▣ **Derivatives:** option prices depend on the volatility expected until maturity
- ▣ **Inference:** tests and confidence intervals that assume a constant variance can be wrong
  - ▶ this is why [Chapter 7](#) needed tests that remain valid under heteroskedasticity

## Stylised Facts a Volatility Model Must Reproduce

- ▣ **Volatility clustering:** the ACF (autocorrelation function) of  $r_t^2$  is positive and decays slowly ([Chapter 8](#))
  - ▶ S&P 500: autocorrelation of  $r_t^2$  at lag 1: 0.31; at lag 50: still 0.053
- ▣ **Heavy tails:** kurtosis above 3, the value of the Normal distribution ([Chapter 2](#))
  - ▶ S&P 500 daily returns: kurtosis 13.7
- ▣ **Leverage effect:** falls raise volatility more than rises of the same size [Christie \(1982\)](#)
- ▣ **Mean reversion of volatility:** after a storm, volatility returns to a long-run level
- ▣ Almost no autocorrelation in  $r_t$  itself ([Chapter 7](#))

## Conditioning on the Past

- $\mathcal{F}_{t-1}$ : the information available at the end of day  $t - 1$  (past returns)
  - ▶ conditional expectation  $E[X \mid \mathcal{F}_{t-1}]$ : the best forecast of  $X$  given the past ([Chapter 4](#))
- **Conditional mean**:  $\mu_t = E[r_t \mid \mathcal{F}_{t-1}]$ 
  - ▶ [Chapter 7](#): for daily returns,  $\mu_t$  is almost constant:  $\mu_t = \mu$
- **Conditional variance**:  $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1}) = E[(r_t - \mu_t)^2 \mid \mathcal{F}_{t-1}]$ 
  - ▶  $\sigma_t$  is the **conditional volatility**: known at  $t - 1$ , it changes from day to day
  - ▶ **heteroskedasticity**: a variance that is not constant

## The Basic Decomposition

- Every model of this chapter has the form  $r_t = \mu + \varepsilon_t$ ,  $\varepsilon_t = \sigma_t z_t$ 
  - ▶  $z_t$ : the **innovations**, i.i.d. with mean 0 and variance 1 (Normal, Student-t, ...)
  - ▶  $\varepsilon_t$ : the **shock** (the return minus its mean);  $\sigma_t$  depends only on the past
- Consequences
  - ▶  $E[\varepsilon_t | \mathcal{F}_{t-1}] = \sigma_t E[z_t] = 0$ : the shocks are a martingale difference, hence uncorrelated
  - ▶  $\text{Var}(\varepsilon_t | \mathcal{F}_{t-1}) = \sigma_t^2$ : the variance is predictable
  - ▶ the  $\varepsilon_t$  are uncorrelated but not independent:  $\varepsilon_t^2$  is correlated with  $\varepsilon_{t-1}^2$
- In the terms of [Chapter 7](#): RW1 (i.i.d. increments) is rejected, RW3 (uncorrelated increments) is kept

## Worked Example: Conditional and Unconditional Variance

- A market has calm days ( $\sigma_t^2 = 0.5$ ) and stormy days ( $\sigma_t^2 = 4.5$ ), each with probability  $1/2$ ; the mean is 0
  - ▶ law of total variance:  $\text{Var}(r_t) = E[\text{Var}(r_t | \mathcal{F}_{t-1})] + \text{Var}(E[r_t | \mathcal{F}_{t-1}])$
  - ▶ step by step:  $0.5 \times 0.5 + 0.5 \times 4.5 + 0 = 2.50$ , so the unconditional standard deviation is 1.58%
- The unconditional variance is an average; the conditional variance tells which day we are in
  - ▶ a calm day:  $\sigma_t = 0.71\%$ ; a stormy day:  $\sigma_t = 2.12\%$ ; the average 1.58% fits neither
- A mixture of Normal distributions with different variances has heavy tails: conditional heteroskedasticity creates kurtosis

## 1982: Robert Engle and ARCH

- **Engle (1982): ARCH** (autoregressive conditional heteroskedasticity): today's variance depends on yesterday's squared shocks
  - ▶ first application: the variance of inflation in the United Kingdom; written while Engle was visiting the London School of Economics
- **Nobel Prize 2003**: "for methods of analyzing economic time series with time-varying volatility (ARCH)"
  - ▶ shared with Clive Granger (cointegration); Nobel lecture: [Engle \(2004\)](#)



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC BY-SA 4.0;

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Stockholm Concert Hall, venue of the Nobel Prize ceremony

Photo: Karen Zhou (2025); CC BY-SA 4.0; [Wikimedia Commons](#)

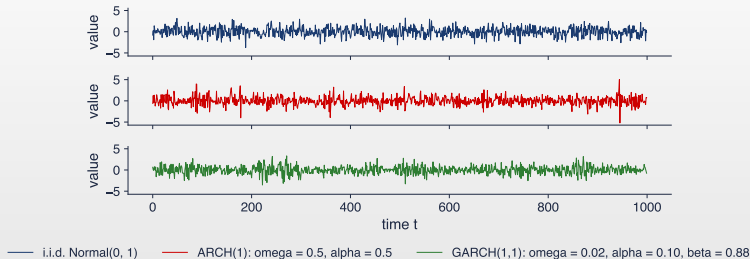
## ARCH(1): Definition

- $\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2, \quad \omega > 0, \alpha \geq 0$ 
  - ▶ in words: today's variance = a constant plus a share  $\alpha$  of yesterday's squared shock
  - ▶  $\omega$ : the variance after a zero shock;  $\alpha$ : the reaction to yesterday's shock; both keep the variance positive
  - ▶ a large shock yesterday, of either sign, raises today's variance
- "Autoregressive":  $\varepsilon_t^2$  follows an AR(1) (autoregressive) model
  - ▶ write  $\varepsilon_t^2 = \sigma_t^2 + v_t$  with  $v_t = \sigma_t^2(z_t^2 - 1)$ ,  $E[v_t | \mathcal{F}_{t-1}] = 0$
  - ▶ then  $\varepsilon_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + v_t$ : an AR(1) for the squared shocks
- Hence the ACF of  $\varepsilon_t^2$  is  $\rho(k) = \alpha^k$  when the fourth moment exists: clustering, but a fast decay

## ARCH(1): Properties

- Mean and autocorrelation:  $E[\varepsilon_t] = 0$ ,  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$  for  $k \neq 0$
- **Unconditional variance** (if  $\alpha < 1$ ): take expectations in  $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$ 
  - ▶  $\bar{\sigma}^2 = E[\varepsilon_t^2] = \omega + \alpha \bar{\sigma}^2$ , so  $\bar{\sigma}^2 = \omega / (1 - \alpha)$
- **Kurtosis** with Normal  $z_t$  (if  $3\alpha^2 < 1$ ):  $K = \frac{E[\varepsilon_t^4]}{(E[\varepsilon_t^2])^2} = 3 \frac{1 - \alpha^2}{1 - 3\alpha^2} > 3$ 
  - ▶ Normal innovations, but heavy-tailed returns [Franke, Härdle & Hafner \(2019\)](#)
  - ▶ if  $\alpha \geq 1/\sqrt{3} = 0.577$ , the fourth moment is infinite
- **Worked example:**  $\omega = 0.5$ ,  $\alpha = 0.5$ 
  - ▶  $\bar{\sigma}^2 = 0.5/0.5 = 1.00$ ;  $K = 3 \times 0.75/0.25 = 9.0$ ; after a shock  $\varepsilon_{t-1} = 2$ :  $\sigma_t^2 = 0.5 + 0.5 \times 4 = 2.50$ ,  $\sigma_t = 1.58$

## Simulated Paths: i.i.d., ARCH(1) and GARCH(1,1)



- Three series of 1000 values with unconditional variance 1; ARCH(1) with  $\omega = \alpha = 0.5$ , GARCH(1,1) with  $\omega = 0.02$ ,  $\alpha = 0.10$ ,  $\beta = 0.88$  (next section)
- Sample kurtosis: i.i.d. 3.12, ARCH 4.91, GARCH 3.74; theory: 3, 9.0, 6.06; the fourth moment converges slowly
- ARCH(1) gives isolated spikes; GARCH(1,1) gives long calm and long agitated periods, as in real returns

## ARCH( $q$ ) and Its Limits

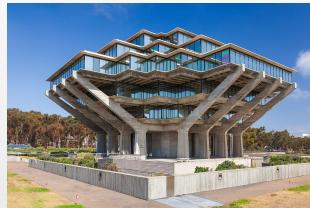
- ▣ **ARCH( $q$ ):**  $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2$ ,  $\alpha_i \geq 0$ ; stationary if  $\sum_i \alpha_i < 1$ 
  - ▶  $q$ : the number of lags;  $\alpha_i$ : the reaction to the shock of  $i$  days ago
  - ▶ the effect of a shock lasts exactly  $q$  days; real clustering lasts for months
- ▣ Testing for ARCH effects: the ARCH-LM test of Engle (regression of  $\hat{\varepsilon}_t^2$  on its lags), [Chapter 8](#)

| S&P 500, Normal $z_t$ | parameters | log-likelihood | AIC            | BIC            | $\sum \alpha_i (+\beta)$ |
|-----------------------|------------|----------------|----------------|----------------|--------------------------|
| ARCH(1)               | 3          | -10305.5       | 20617.0        | 20637.5        | 0.394                    |
| ARCH(5)               | 7          | -9340.5        | 18695.1        | 18742.7        | 0.816                    |
| ARCH(10)              | 12         | -9227.6        | 18479.3        | 18561.0        | 0.849                    |
| GARCH(1,1)            | 4          | -9218.0        | <b>18444.0</b> | <b>18471.2</b> | 0.980                    |

- ▣ Interpretation: ARCH needs ten lags and still loses to GARCH(1,1), which has four parameters (AIC and BIC: smaller is better, Section 10)

## 1986: Tim Bollerslev and GARCH

- [Bollerslev \(1986\)](#), a doctoral student of Engle at the University of California, San Diego: add yesterday's variance to the equation
  - ▶ **GARCH** (generalised ARCH): a few parameters replace a long ARCH( $q$ )
- Student-t innovations for returns: [Bollerslev \(1987\)](#)
- GARCH(1,1) is still the benchmark: [Hansen & Lunde \(2005\)](#) compared 330 models and found it hard to beat on exchange rates



Geisel Library, University of California, San Diego

Photo: [Stephen Bay \(2010\)](#); CC BY 4.0; [Wikimedia](#)

[Commons](#)

## GARCH(1,1): Definition

- $r_t = \mu + \varepsilon_t$ ,  $\varepsilon_t = \sigma_t z_t$ ,  $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ 
  - ▶  $\omega > 0$ ,  $\alpha \geq 0$ ,  $\beta \geq 0$ : the variance stays positive
- Reading the parameters
  - ▶  $\alpha$ : the **reaction** to yesterday's news (size of the jump after a shock)
  - ▶  $\beta$ : the **memory** of the variance (how much of yesterday's level is kept)
  - ▶  $\alpha + \beta$ : the **persistence**;  $\omega$  sets the long-run level
- By recursion, an ARCH( $\infty$ ) with geometric weights
  - ▶  $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j=0}^{\infty} \beta^j \varepsilon_{t-1-j}^2$ : all past shocks matter, with decaying weights
  - ▶ the shock of  $j+1$  days ago gets the weight  $\alpha\beta^j$

## Stationarity, Long-Run Variance and Kurtosis

□ As for ARCH:  $\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + v_t - \beta v_{t-1}$ , an ARMA(1,1) for the squared shocks

- ▶  $v_t = \varepsilon_t^2 - \sigma_t^2$ : the surprise in the squared shock, as for ARCH(1)
- ▶ ARMA: autoregressive moving average; the ACF of  $\varepsilon_t^2$  decays like  $(\alpha + \beta)^k$  at lag  $k$

□ **Covariance stationary** if  $\alpha + \beta < 1$ ; then the **unconditional (long-run) variance** is

- ▶  $\bar{\sigma}^2 = E[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$ , annualised volatility  $\sqrt{252 \bar{\sigma}^2}$  for 252 trading days

□ Kurtosis with Normal  $z_t$ , if  $(\alpha + \beta)^2 + 2\alpha^2 < 1$ :  $K = 3 \frac{1 - (\alpha + \beta)^2}{1 - (\alpha + \beta)^2 - 2\alpha^2} > 3$

- ▶  $\alpha = 0.10, \beta = 0.88$ :  $(\alpha + \beta)^2 + 2\alpha^2 = 0.9804$ ,  $K = 6.06$

## Persistence and Half-Life

- ▣ Forecast of the variance  $h$  days ahead (derived in Section 11):  $E[\sigma_{t+h}^2 | \mathcal{F}_t] - \bar{\sigma}^2 = (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$ 
  - ▶  $E[\cdot | \mathcal{F}_t]$ : the forecast made at the end of day  $t$ ;  $\bar{\sigma}^2$ : the long-run variance
  - ▶ a deviation from the long-run variance shrinks by the factor  $\alpha + \beta$  each day
- ▣ **Half-life**: the number of days after which half of a variance shock is gone
  - ▶  $(\alpha + \beta)^h = 1/2 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
  - ▶  $\alpha + \beta = 0.90$ : 6.6 days; 0.98: 34.3 days; 0.99: 69.0 days: small changes near 1 matter a lot
- ▣ Daily equity returns:  $\alpha$  around 0.05–0.15,  $\beta$  around 0.80–0.93,  $\alpha + \beta$  close to 1 (Section 7)

## Worked Example: One GARCH(1,1) Step

- ▣ Daily returns in %:  $\omega = 0.02$ ,  $\alpha = 0.10$ ,  $\beta = 0.88$ ,  $\mu = 0$
- ▣ Long-run quantities
  - ▶ persistence  $\alpha + \beta = 0.98$ ;  $\bar{\sigma}^2 = 0.02 / (1 - 0.98) = 1.00$ ; annualised volatility  $\sqrt{252 \times 1.00} = 15.9\%$
  - ▶ half-life  $\ln 0.5 / \ln 0.98 = 34.3$  days
- ▣ Today:  $\sigma_t^2 = 1.2$  and the return is  $r_t = -3\%$ 
  - ▶ tomorrow:  $\sigma_{t+1}^2 = 0.02 + 0.10 \times 9 + 0.88 \times 1.2 = 1.976$ ,  $\sigma_{t+1} = 1.41\%$
  - ▶ the shock raised the variance from 1.2 to almost 2; it then decays towards 1.00 with the factor 0.98 per day

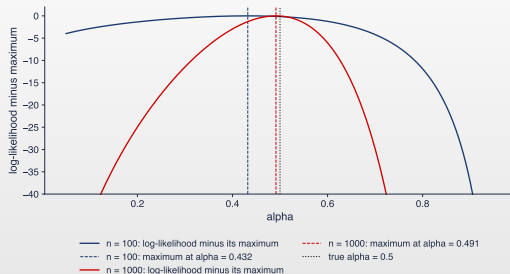
## IGARCH and the EWMA Link

- ▣ **IGARCH** (integrated GARCH) Engle & Bollerslev (1986):  $\alpha + \beta = 1$ 
  - ▶ shocks to the variance never die out: no half-life, no finite unconditional variance
  - ▶ the forecast of the variance is the same for every horizon
- ▣ **EWMA** (exponentially weighted moving average, Chapter 8):  $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$ 
  - ▶ this is IGARCH with  $\omega = 0$ ,  $\alpha = 1 - \lambda$ ,  $\beta = \lambda$ ; RiskMetrics uses  $\lambda = 0.94$  for daily data
  - ▶ EWMA: nothing to estimate, but no mean reversion; GARCH: three parameters, and volatility returns to  $\bar{\sigma}$
- ▣ **Question for the room:** an analyst forecasts the volatility one year ahead with EWMA right after a crash. What goes wrong?
  - ▶ **Answer:** EWMA keeps today's crisis level for the whole year; GARCH lets it decay towards the long-run level

## The Likelihood of a GARCH Model

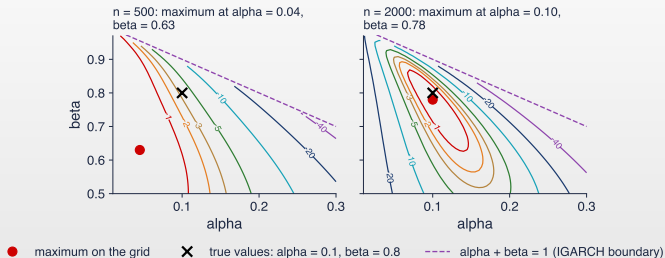
- The returns are dependent, so we factor the joint density into conditional densities
  - ▶  $f$ : a probability density;  $T$ : the number of returns;  $\theta$ : the vector of parameters;  $\ell$ : the log-likelihood
  - ▶  $f(r_1, \dots, r_T) = f(r_1) \prod_{t=2}^T f(r_t | \mathcal{F}_{t-1})$ , and  $r_t | \mathcal{F}_{t-1} \sim N(\mu, \sigma_t^2)$  for Normal innovations
- **Conditional log-likelihood** with  $\theta = (\mu, \omega, \alpha, \beta)$ :
  - ▶  $\ell(\theta) = -\frac{1}{2} \sum_{t=2}^T \left[ \ln(2\pi) + \ln \sigma_t^2(\theta) + \frac{(r_t - \mu)^2}{\sigma_t^2(\theta)} \right]$
  - ▶ in words: each day adds a penalty for a large  $\sigma_t^2$  and a penalty for a squared error that is large relative to  $\sigma_t^2$
  - ▶  $\sigma_t^2(\theta)$  is computed recursively from a starting value  $\sigma_1^2$  (for example the sample variance)
- **MLE** (maximum likelihood estimation):  $\hat{\theta} = \arg \max_{\theta} \ell(\theta)$ , under  $\omega > 0$ ,  $\alpha, \beta \geq 0$ ,  $\alpha + \beta < 1$ 
  - ▶ no closed form: a numerical optimiser searches for the maximum [Franke, Härdle & Hafner \(2019\)](#)

## The Likelihood of ARCH(1)



- ▣ Simulated ARCH(1) with  $\omega = \alpha = 0.5$ ;  $\ell(\alpha)$  with  $\omega = 1 - \alpha$ , as in SFElkarch1; each curve minus its maximum
- ▣  $n = 100$ :  $\hat{\alpha} = 0.432$  (standard error 0.127);  $n = 1000$ :  $\hat{\alpha} = 0.491$  (0.035)
- ▣ Interpretation: more data make the curve sharper; its curvature at the maximum gives the standard error

## The Likelihood of GARCH(1,1)



- Simulated GARCH(1,1),  $\omega = 0.1$ ,  $\alpha = 0.1$ ,  $\beta = 0.8$ ;  $\omega$  set so that the unconditional variance equals the sample variance, as in SFEligarch; contours: log-likelihood minus its maximum
- $n = 500$ : maximum at (0.04, 0.63), far from the truth;  $n = 2000$ : (0.10, 0.78)
- Interpretation: a long, flat ridge along which  $\alpha$  and  $\beta$  trade off: GARCH needs long samples

## Estimation Step by Step and with a Package

| S&P 500, GARCH(1,1)-N | step by step (SE) | arch   | classic SE | robust SE |
|-----------------------|-------------------|--------|------------|-----------|
| $\mu$                 | 0.0614 (0.0098)   | 0.0615 | 0.0098     | 0.0100    |
| $\omega$              | 0.0253 (0.0030)   | 0.0254 | 0.0030     | 0.0048    |
| $\alpha$              | 0.1205 (0.0086)   | 0.1206 | 0.0086     | 0.0118    |
| $\beta$               | 0.8600 (0.0091)   | 0.8599 | 0.0091     | 0.0125    |

- Step by step: write  $-\ell(\theta)$  in Python and minimise it with `scipy.optimize.minimize` under  $\alpha + \beta < 1$ 
  - the two routes differ only in the start-up of  $\sigma_t^2$  and in the optimiser; log-likelihood  $-9219.9$  against  $-9218.0$
- SE (standard error), classic: from the inverse Hessian (the curvature of  $\ell$ ); robust: valid when  $z_t$  is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- Interpretation: the robust SE are about 1.4 times larger (1.6 for  $\omega$ ): with heavy-tailed  $z_t$  the classic SE overstate the precision

## Quasi-Maximum Likelihood

- ▣ **QMLE** (quasi-maximum likelihood estimation): maximise the Normal likelihood even if  $z_t$  is not Normal
  - ▶ the estimates are still consistent if  $\mu$  and  $\sigma_t^2$  are correctly specified [Bollerslev & Wooldridge \(1992\)](#)
  - ▶ but the classic SE are wrong: use the robust ("sandwich") SE
- ▣ Practical rules
  - ▶ report robust SE (the default of `arch`)
  - ▶ use at least 1000–2000 daily observations; returns in % avoid numerical problems with tiny  $\omega$
  - ▶ an estimate on the boundary ( $\alpha + \beta = 1$ ) has no usual standard error: report it as IGARCH
- ▣ A better innovation distribution gives more efficient estimates: next section

## Student-t and Skewed-t Innovations

- ▣ GARCH with Normal  $z_t$  explains only part of the kurtosis: S&P 500 returns 13.7, standardised residuals  $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ : 4.7
  - ▶ the remaining heavy tails must come from the distribution of  $z_t$
- ▣ **Standardised Student-t** with  $\nu > 2$  degrees of freedom ([Chapter 2](#)) [Bollerslev \(1987\)](#)
  - ▶  $t_\nu$ : a Student-t variable with  $\nu$  degrees of freedom;  $z_t = t_\nu \sqrt{(\nu - 2)/\nu}$  has variance 1
  - ▶ a small  $\nu$  means heavy tails: kurtosis  $3 + 6/(\nu - 4)$  for  $\nu > 4$ ;  $\nu \rightarrow \infty$ : the Normal distribution
  - ▶  $\nu$  is estimated together with  $(\mu, \omega, \alpha, \beta)$
- ▣ **Skewed t** [Hansen \(1994\)](#): one more parameter  $\lambda \in (-1, 1)$ ;  $\lambda < 0$ : a longer left tail
  - ▶ nested models: Normal  $\subset$  t  $\subset$  skewed t; compare them with likelihood-ratio tests or AIC/BIC ([Chapter 6](#))

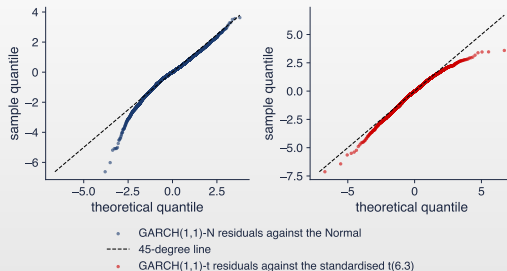
## S&P 500: Three Innovation Distributions

| GARCH(1,1) | $\omega$ | $\alpha$ | $\beta$ | $\nu$ | $\lambda$ | log-lik. | $\alpha + \beta$ |
|------------|----------|----------|---------|-------|-----------|----------|------------------|
| Normal     | 0.0254   | 0.121    | 0.860   | —     | —         | −9218.0  | 0.980            |
| Student-t  | 0.0157   | 0.123    | 0.872   | 6.26  | —         | −9062.5  | 0.995            |
| skewed t   | 0.0153   | 0.120    | 0.872   | 6.92  | −0.114    | −9038.4  | 0.993            |

- Likelihood-ratio statistic  $2(\ell_1 - \ell_0)$  ( $\ell_1, \ell_0$ : maximised log-likelihoods of the larger and the smaller model): t against Normal 311.0 (1 restriction,  $\chi^2_{0.95}(1) = 3.84$ ); skewed t against t 48.1: both reject the smaller model
- $\hat{\nu} = 6.26$ : tails far heavier than Normal;  $\hat{\lambda} = -0.114$ : large falls more frequent than large rises
- Interpretation:  $\alpha$  and  $\beta$  hardly change; the innovation distribution matters for tail quantiles (VaR), less for  $\sigma_t$

 SFM\_ch9\_garch\_estimation

## QQ Plots of the Standardised Residuals



- Left: residuals of the Normal GARCH against the Normal distribution: the 0.1% quantile is  $-4.81$ , against  $-3.09$  in theory
- Right: residuals of GARCH-t against the standardised t with  $\hat{\nu} = 6.3$ :  $-4.79$  against  $-4.19$ : much closer; the largest falls are still deeper
- Interpretation: the left tail is the problem: a reason for skewed innovations and for asymmetric models (Section 8)

## GARCH(1,1)-t Estimates

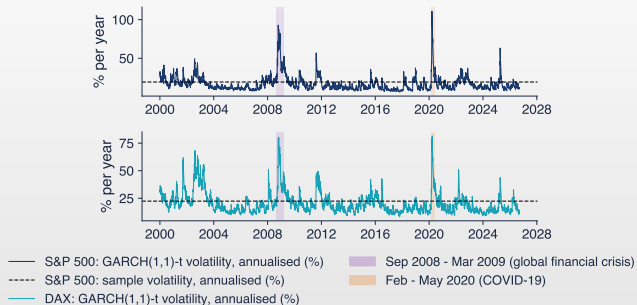
|         | from | $\alpha$ | $\beta$ | $\nu$ | $\alpha + \beta$ | half-life | long-run vol. | sample vol. |
|---------|------|----------|---------|-------|------------------|-----------|---------------|-------------|
| S&P 500 | 2000 | 0.123    | 0.872   | 6.26  | 0.995            | 131       | 27.3          | 19.2        |
| DAX     | 2000 | 0.097    | 0.895   | 7.08  | 0.992            | 88        | 25.9          | 22.4        |
| BET     | 2000 | 0.192    | 0.799   | 5.01  | 0.991            | 74        | 34.7          | 22.1        |
| Bitcoin | 2014 | 0.109    | 0.891   | 3.18  | 1.000            | —         | —             | 66.6        |
| TLV     | 2010 | 0.139    | 0.824   | 4.91  | 0.963            | 18        | 28.2          | 26.5        |
| SNP     | 2010 | 0.135    | 0.807   | 4.81  | 0.942            | 12        | 25.6          | 24.5        |

- Daily log returns in % to 18 September 2026; robust SE of  $\hat{\alpha}$  and  $\hat{\beta}$  between 0.010 and 0.041; half-life in days; volatilities annualised with the actual number of observations per year, in %
- Indices:  $\alpha + \beta$  between 0.991 and 0.995, half-lives of 74–131 trading days; BVB stocks: shorter memory (TLV 18 days, SNP 12 days)
- BET: the largest  $\alpha$  (0.192): a stronger reaction to news; all  $\hat{\nu}$  between 3.2 and 7.1: heavy tails everywhere

## Interpretation: Bitcoin and the Long-Run Volatility

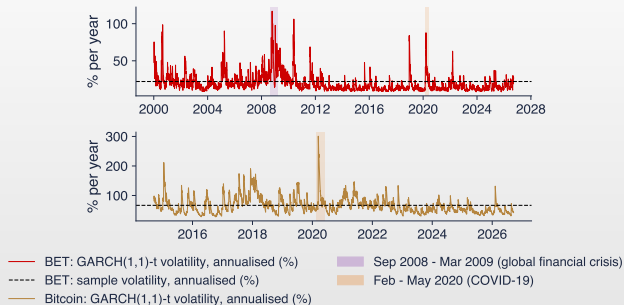
- Bitcoin:  $\hat{\alpha} + \hat{\beta} = 1.0000$ , on the boundary  $\alpha + \beta < 1$ : an IGARCH
  - ▶ no half-life and no long-run volatility: the model behaves like an EWMA with  $\lambda \approx 0.891$
  - ▶  $\hat{\nu} = 3.18$ : extremely heavy tails (the fourth moment of  $z_t$  does not exist)
- **Question for the room:** S&P 500: long-run volatility 27.3%, sample volatility 19.2%. Is the model wrong?
  - ▶ **Answer:** not necessarily:  $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$  divides by  $1 - \alpha - \beta = 0.0053$ , a small and imprecise number
  - ▶ the Normal GARCH gives 18.1%; the long-run level is the least reliable output of a persistent GARCH

## Conditional Volatility: S&P 500 and DAX



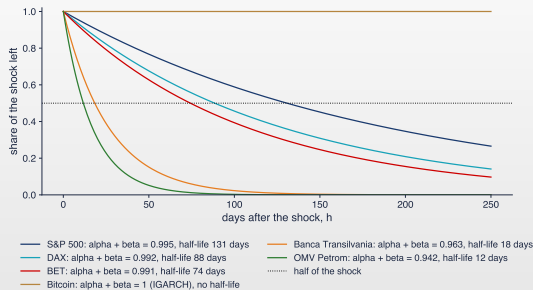
- GARCH(1,1)-t volatility  $\hat{\sigma}_t$ , annualised, as in SFEvolgarchest; dashed: the sample volatility
- S&P 500: peaks 93% (16 October 2008) and 111% (17 March 2020); median 14%; on 18 September 2026: 12%

## Conditional Volatility: BET and Bitcoin



- BET: the highest peak in 2008 (117%, 15 October 2008), lower in 2020 (88%); Bitcoin: 301% on 13 March 2020, median 59%
- Interpretation: the 2020 shock hit all markets in the same week; the median volatility of Bitcoin is 4.1 times that of the S&P 500 (14%)

## How Long Does a Shock Last?



- Share of a variance shock left after  $h$  days:  $(\alpha + \beta)^h$ ; the half-life is where the curve crosses  $1/2$
- Interpretation: after a crisis, the S&P 500 needs about half a year to halve the excess variance; OMV Petrom about two weeks; Bitcoin never

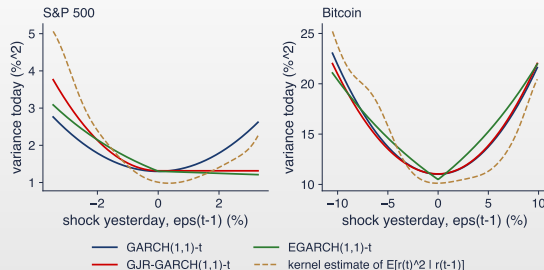
## The Leverage Effect

- ▣ The leverage effect of Section 1 [Christie \(1982\)](#) has two explanations
  - ▶ **leverage** explanation: a lower share price raises the debt-to-equity ratio, so the equity becomes riskier
  - ▶ **volatility feedback**: news of higher risk lowers prices at once
- ▣ GARCH(1,1) cannot see it:  $\sigma_t^2$  depends on  $\varepsilon_{t-1}^2$ , not on the sign of  $\varepsilon_{t-1}$ 
  - ▶ the QQ plot of Section 6 and the skewed t ( $\hat{\lambda} < 0$ ) already pointed to the left tail
- ▣ **NIC** (news impact curve) [Engle & Ng \(1993\)](#):  $\sigma_t^2$  as a function of  $\varepsilon_{t-1}$ , with  $\sigma_{t-1}^2$  fixed at the unconditional variance

## GJR-GARCH and EGARCH

- **GJR-GARCH** [Glosten, Jagannathan & Runkle \(1993\)](#):  $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ ,  $I_{t-1} = 1$  if  $\varepsilon_{t-1} < 0$ , else 0
  - ▶  $I_{t-1}$ : an indicator of a negative shock yesterday;  $\gamma$ : the extra reaction to bad news
  - ▶ good news: slope  $\alpha$ ; bad news:  $\alpha + \gamma$ ; leverage effect:  $\gamma > 0$
  - ▶ persistence  $\alpha + \beta + \gamma/2$  for symmetric  $z_t$  (half of the shocks are negative)
- **EGARCH** (exponential GARCH) [Nelson \(1991\)](#):  $\ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ 
  - ▶  $|z_{t-1}| - E|z_{t-1}|$ : the size of yesterday's standardised shock above its average ( $E|z| = \sqrt{2/\pi}$  for Normal  $z$ )
  - ▶ the logarithm keeps  $\sigma_t^2 > 0$  without sign restrictions on the parameters
  - ▶ leverage effect:  $\gamma < 0$  (a negative  $z_{t-1}$  raises  $\ln \sigma_t^2$ ); persistence:  $\beta$
- Textbook treatment of both, and of the threshold GARCH: [Franke, Härdle & Hafner \(2019\)](#), Section 13.2

## News Impact Curves: S&P 500 and Bitcoin



- Model curves:  $\sigma_t^2$  against  $\varepsilon_{t-1}$ ,  $\sigma_{t-1}^2$  fixed at the sample variance; dashed: a kernel (Nadaraya–Watson) estimate of  $E[r_t^2 | r_{t-1}]$ , the idea of SFNewsImpactCurve
- S&P 500, GJR: after a fall of 2%, tomorrow's variance is 1.62 times the variance after a rise of 2%; Bitcoin: ratio 1.00, a symmetric curve

## Asymmetry in Six Markets

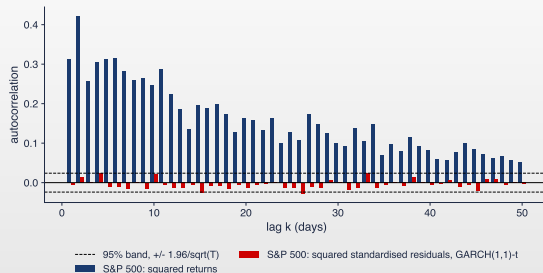
|         | GJR $\alpha$ | GJR $\gamma$ | $t(\gamma)$ | LR    | p       | EGARCH $\gamma$ | $t(\gamma)$ |
|---------|--------------|--------------|-------------|-------|---------|-----------------|-------------|
| S&P 500 | 0.000        | 0.205        | 10.0        | 235.7 | < 0.001 | -0.165          | -15.2       |
| DAX     | 0.000        | 0.163        | 8.7         | 204.5 | < 0.001 | -0.133          | -11.8       |
| BET     | 0.172        | 0.043        | 2.1         | 4.9   | 0.026   | -0.028          | -2.6        |
| Bitcoin | 0.113        | -0.013       | -0.7        | 0.7   | 0.416   | 0.015           | 1.2         |
| TLV     | 0.101        | 0.065        | 2.7         | 7.8   | 0.005   | -0.040          | -3.0        |
| SNP     | 0.117        | 0.033        | 1.4         | 1.7   | 0.188   | -0.027          | -1.9        |

- Student-t innovations;  $t$  statistics with robust SE; LR (likelihood ratio) =  $2(\ell_{GJR} - \ell_{GARCH}) \sim \chi^2(1)$
- S&P 500 and DAX: GJR  $\hat{\alpha} = 0$ : only bad news raises volatility; a very strong leverage effect
- BET and TLV: a weak effect, significant by LR, yet BIC prefers the symmetric GARCH; SNP: not significant at 5%; Bitcoin: none (EGARCH  $\gamma$  slightly positive, not significant)
- Interpretation: leverage is a property of mature equity markets; in crypto, large rises are as “frightening” as large falls

## Checking a Fitted Model

- If the model is right, the **standardised residuals**  $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$  are close to i.i.d. with mean 0 and variance 1
  - ▶ **LB** (Ljung–Box)  $Q(10)$  on  $\hat{z}_t$ : no autocorrelation left in the mean ([Chapter 7](#)) [Ljung & Box \(1978\)](#)
  - ▶  $Q(10)$  on  $\hat{z}_t^2$  and the ARCH-LM test: no ARCH effects left ([Chapter 8](#))
  - ▶ QQ plot of  $\hat{z}_t$ : is the assumed distribution right? (Section 6)
- **Sign-bias test** [Engle & Ng \(1993\)](#): regress  $\hat{z}_t^2$  on  $I_{t-1}$ ,  $I_{t-1}\hat{\varepsilon}_{t-1}$  and  $(1 - I_{t-1})\hat{\varepsilon}_{t-1}$ 
  - ▶ joint test  $TR^2 \sim \chi^2(3)$ : a rejection means the sign of past shocks still predicts the variance (asymmetry is missing)
- A passed diagnostic does not prove the model right; a failed one shows where it is wrong

## What GARCH Removes



- ACF of  $r_t^2$ : 0.31 at lag 1 and still 0.053 at lag 50; ACF of  $\hat{z}_t^2$  (GARCH(1,1)-t):  $-0.005$  at lag 1, at most 0.028 in absolute value; band  $\pm 0.024$
- Interpretation: three parameters absorb almost all the volatility clustering in the daily data since 2000

## Diagnostics on Real Data

| S&P 500                    | $Q(10)$ (p)    | $Q(10)$ of squares (p) | ARCH-LM(5) (p) | sign bias $\chi^2(3)$ (p) |
|----------------------------|----------------|------------------------|----------------|---------------------------|
| returns $r_t$              | 96.3 (< 0.001) | 6133 (< 0.001)         | 1679 (< 0.001) | –                         |
| $\hat{z}_t$ , Normal GARCH | 21.2 (0.020)   | 16.4 (0.089)           | 8.0 (0.154)    | –                         |
| $\hat{z}_t$ , GARCH-t      | 21.9 (0.016)   | 13.6 (0.192)           | 5.8 (0.326)    | 40.7 (< 0.001)            |

- GARCH removes the ARCH effects ( $Q(10)$  of squares: from 6133 to 13.6); a little autocorrelation in the mean remains (constant-mean model)
- The sign-bias test rejects for the symmetric GARCH (sign  $t = 3.56$ ): the asymmetry of Section 8 is missing; similar for the DAX ( $p < 0.001$ ) and the BET ( $p = 0.009$ )
- Other markets,  $Q(10)$  of  $\hat{z}_t^2$  (GARCH-t):  $p$  between 0.115 (DAX) and 0.993 (TLV): no ARCH effect left anywhere

## Nine Models for the S&P 500

- **AIC** (Akaike information criterion) =  $-2\ell + 2k$  Akaike (1974); **BIC** (Bayesian information criterion) =  $-2\ell + k \ln T$  Schwarz (1978);  $\ell$ : maximised log-likelihood,  $k$  parameters,  $T$  observations; smaller is better (Chapter 6)

|                | Normal   |              | Student-t |              | skewed t |              |
|----------------|----------|--------------|-----------|--------------|----------|--------------|
|                | log-lik. | $\Delta$ BIC | log-lik.  | $\Delta$ BIC | log-lik. | $\Delta$ BIC |
| GARCH(1,1)     | -9218.0  | 648.9        | -9062.5   | 346.7        | -9038.4  | 307.4        |
| GJR-GARCH(1,1) | -9086.5  | 394.7        | -8944.6   | 119.9        | -8905.1  | 49.6         |
| EGARCH(1,1)    | -9075.6  | 372.9        | -8922.1   | 74.8         | -8880.3  | 0.0          |

- $\Delta$ BIC: the difference from the best model; AIC gives the same ranking
- Both choices matter: starting from the Normal GARCH, asymmetry alone (Normal EGARCH) lowers BIC by 276, heavy tails alone (GARCH-skewed t) by 341, both together by 649
- Interpretation: EGARCH with skewed-t innovations wins; with 6 714 observations even small improvements are “significant”; the forecast comparison in Section 11 is the real test

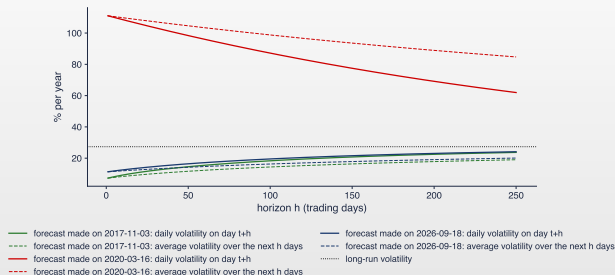
## Multi-Step Forecasts

- One step:  $\sigma_{t+1}^2 = \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2$  is known at the end of day  $t$ 
  - ▶ two steps:  $E_t[\sigma_{t+2}^2] = \omega + \alpha E_t[\varepsilon_{t+1}^2] + \beta \sigma_{t+1}^2 = \omega + (\alpha + \beta) \sigma_{t+1}^2$ , since  $E_t[\varepsilon_{t+1}^2] = \sigma_{t+1}^2$
- By induction:  $E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$ 
  - ▶  $E_t$ : the expectation given the information at the end of day  $t$ ;  $h$ : the horizon in days
  - ▶ the forecast converges to the long-run variance; the speed is set by  $\alpha + \beta$
- Variance of the  $H$ -day return: the sum of the daily forecasts (returns are uncorrelated)
  - ▶  $\text{Var}_t(r_{t+1} + \dots + r_{t+H}) = \sum_{h=1}^H E_t[\sigma_{t+h}^2]$ , not  $H\sigma_{t+1}^2$
  - ▶ the square-root-of-time rule  $\sigma_{t+1}\sqrt{H}$  is too high in a crisis and too low in a calm period

## Worked Example: the S&P 500 on 18 September 2026

- ▣ GARCH(1,1)-t:  $\alpha + \beta = 0.9947$ ,  $\bar{\sigma}^2 = 2.972$ ; forecast for the next day  $\sigma_{t+1}^2 = 0.509$  (below the long-run level)
- ▣ Step by step
  - ▶ day 10:  $2.972 + 0.9947^9(0.509 - 2.972) = 2.972 + 0.953 \times (0.509 - 2.972) = 0.624$
  - ▶ 10-day variance: the sum of the ten daily forecasts = 5.67, so the 10-day volatility is 2.38%
  - ▶ square-root-of-time rule:  $\sqrt{10 \times 0.509} = 2.26\%$ : too low, because volatility is expected to rise towards its long-run level

## The Term Structure of Volatility



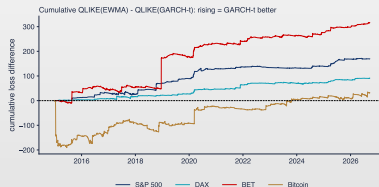
- S&P 500, GARCH(1,1)-t estimated on all data; forecasts made on a calm day (3 November 2017), at the COVID-19 peak (16 March 2020) and on 18 September 2026
- Calm day: 7.3% tomorrow, average 19.0% over a year; COVID-19 peak: 111.0% tomorrow, still 84.7% on average over a year
- Interpretation: all curves move towards the long-run level (27.3%) at the same slow speed: the slope of the term structure tells whether we are in a calm or a storm

## How to Evaluate a Volatility Forecast

- The true  $\sigma_t^2$  is never observed: we compare the forecast  $h_t$  with a noisy **proxy**, here  $r_t^2$ 
  - ▶  $E_{t-1}[r_t^2] = \sigma_t^2$  (with  $\mu \approx 0$ ), but  $r_t^2$  is very noisy: low  $R^2$  even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
- **QLIKE** loss [Patton \(2011\)](#):  $L(r_t^2, h_t) = \frac{r_t^2}{h_t} + \ln h_t$  (up to a constant); smaller is better
  - ▶  $h_t$ : the variance forecast for day  $t$ , made at the end of day  $t - 1$
  - ▶ ranks forecasts correctly even with a noisy proxy; it is minus the Normal log-likelihood
  - ▶ penalises forecasts that are too low more than forecasts that are too high
- **Out of sample**: from 2015, each forecast uses only data up to the day before; the models are re-estimated every 250 days
  - ▶ the **DM** (Diebold–Mariano) test [Diebold & Mariano \(1995\)](#): is the mean loss difference zero?  $t$  with HAC (heteroskedasticity and autocorrelation consistent) standard errors

## GARCH against EWMA, Out of Sample

| 2015–2026 | days  | QLIKE GARCH-t | QLIKE GJR-t | QLIKE EWMA | DM $t$ , GARCH-t vs EWMA (p) |
|-----------|-------|---------------|-------------|------------|------------------------------|
| S&P 500   | 2 944 | 0.761         | 0.733       | 0.819      | −3.69 (< 0.001)              |
| DAX       | 2 973 | 1.090         | 1.064       | 1.121      | −3.19 (0.001)                |
| BET       | 2 926 | 0.680         | 0.674       | 0.788      | −2.22 (0.027)                |
| Bitcoin   | 4 279 | 3.383         | 3.401       | 3.390      | −0.20 (0.845)                |

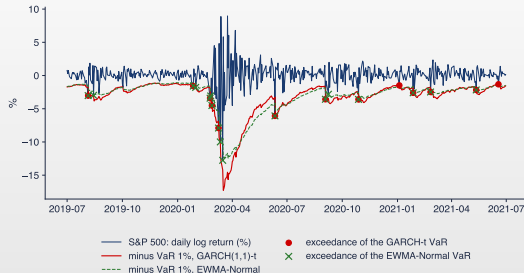


- Equity indices: GARCH-t beats EWMA ( $t < -2$ ); GJR-t is better still for the S&P 500, DAX and BET
- Interpretation: Bitcoin: no difference ( $t = -0.20$ ): its GARCH is an IGARCH, which is almost an EWMA (Section 7)

## Conditional VaR

- **VaR** at level  $\alpha$  (value at risk): the loss exceeded with probability  $\alpha$ :  $\text{VaR}_\alpha = -q_\alpha(r_{t+1})$  (Chapter 6)
  - ▶ here  $\alpha$  is the VaR level, not the GARCH parameter;  $q_\alpha$ : the  $\alpha$ -quantile
  - ▶ here  $\alpha = 1\%$ : VaR 1%, a loss exceeded on one day in a hundred
- With  $r_{t+1} = \mu + \sigma_{t+1}z_{t+1}$ :  $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1} q_\alpha(z))$ 
  - ▶ Normal:  $q_{0.01} = -2.326$ ; standardised t:  $q_{0.01} = t_\nu^{-1}(0.01)\sqrt{(\nu - 2)/\nu}$
  - ▶  $t_\nu^{-1}(0.01)$ : the 1% quantile of the Student-t with  $\nu$  degrees of freedom; the square root rescales it to variance 1
  - ▶ the VaR moves every day with  $\sigma_{t+1}$ : high in storms, low in calm periods
- **Worked example:** S&P 500 on 18 September 2026, GARCH(1,1)-t:  $\hat{\mu} = 0.079$ ,  $\sigma_{t+1} = 0.714$ ,  $\hat{\nu} = 6.26$ 
  - ▶  $q_{0.01}(z) = -3.100 \times 0.825 = -2.557$ ; VaR 1% =  $-(0.079 + 0.714 \times (-2.557)) = 1.75\%$
  - ▶ with Normal  $z_t$ : 1.58%; over 10 days (sum of variances,  $\mu$  ignored):  $2.557 \times \sqrt{5.67}$ , about 6.09%

## VaR 1% through the COVID-19 Crash



- S&P 500, July 2019 – June 2021 (505 days): one-day VaR 1% from GARCH(1,1)-t (re-estimated every 250 days) and from EWMA with Normal  $z_t$
- Exceedances (return below  $-\text{VaR}$ ): GARCH-t 13, EWMA-Normal 17; expected about 5; both models react, but only after the first large falls of March 2020

## Exceedances 2015–2026: a Preview of Backtesting

| VaR 1%  | days  | expected | GARCH-t    | EWMA-Normal |
|---------|-------|----------|------------|-------------|
| S&P 500 | 2 944 | 29       | 49 (1.66%) | 65 (2.21%)  |
| DAX     | 2 973 | 30       | 45 (1.51%) | 74 (2.49%)  |
| BET     | 2 926 | 29       | 34 (1.16%) | 67 (2.29%)  |
| Bitcoin | 4 279 | 43       | 64 (1.50%) | 93 (2.17%)  |

- A correct VaR 1% is exceeded on 1% of the days, and the exceedances do not cluster
- EWMA-Normal: 2.2–2.5%: the Normal quantile is too small for heavy tails
- GARCH-t: 1.2–1.7%: much closer to 1%, but still above it for the S&P 500 and the DAX (the leverage effect is missing)
- Are these differences significant? [Chapter 10](#): the test of [Kupiec \(1995\)](#), ES 2.5%, backtesting
- Further reading on VaR forecasting for Bitcoin: [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

## An Open Question

- **Is Bitcoin's volatility becoming more like the volatility of an equity index?**
  - ▶ today: IGARCH,  $\hat{\nu} = 3.18$ , no leverage effect; the S&P 500: mean-reverting, strong leverage
  - ▶ since 2024, spot Bitcoin ETFs (exchange-traded funds) bring equity investors to the crypto market
- Earlier evidence on crypto assets as an asset class: [Pele et al. \(2023\)](#); volatility as information: [Pele, Lazar & Dufour \(2017\)](#)
- Why it is open: regime changes, the 2022 crypto crash and the ETF launch overlap; the answer depends on the window
- AI tools can speed up such a study, but its results must still be checked [Wang et al. \(2023\)](#)

## How AI Could Help

- ▣ **Literature:** a list of studies on GARCH models for crypto assets and the models they use
- ▣ **Code:** a first draft of rolling GJR-GARCH-t estimates for Bitcoin and the S&P 500 with the arch package
- ▣ **Robustness:** other windows, Ethereum, weekend days removed, skewed-t innovations
- ▣ Example prompt
  - ▶ Write Python code that estimates a GJR-GARCH(1,1) model with Student-t innovations on daily Bitcoin and S&P 500 log returns in percent, on rolling windows of 750 days moved by 21 days, and plots the persistence, gamma and nu over time.

## What to Check

- ▣ Units: returns in % (not decimals), otherwise the optimiser may stop at a wrong  $\omega$
- ▣ The persistence of GJR is  $\alpha + \beta + \gamma/2$ , not  $\alpha + \beta$ ; for EGARCH it is  $\beta$
- ▣ Estimates on the boundary ( $\alpha + \beta = 1$ ): no half-life, no long-run volatility; do not report infinite numbers
- ▣ Calendars: Bitcoin trades 7 days a week, the S&P 500 five; annualise each series with its own frequency
- ▣ Rolling windows overlap: consecutive estimates are not independent evidence
- ▣ References: every cited paper must exist; check the DOI

## Project Idea

- **Question:** has Bitcoin's volatility moved closer to equity volatility since 2021?
  - ▶ data: Bitcoin since 2014, Ethereum since 2015, S&P 500 and BET (EODHD)
- Steps
  - ▶ GJR-GARCH-t on 2015–2019 and on 2021–2026: persistence,  $\gamma$ ,  $\nu$
  - ▶ correlation of the Bitcoin and S&P 500 GARCH volatilities in each period
  - ▶ out-of-sample QLIKE of GARCH against EWMA in each period
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

## Key Takeaways

- ▣ Returns: unpredictable mean, predictable variance:  $r_t = \mu + \sigma_t z_t$
- ▣ GARCH(1,1) with three parameters captures volatility clustering in all our series;  $\alpha + \beta$  is close to 1
- ▣ Estimate by maximum likelihood, report robust SE; Student-t or skewed-t innovations for the tails
- ▣ Equity indices need asymmetry (GJR, EGARCH); Bitcoin does not, and is an IGARCH
- ▣ Forecasts revert to the long-run level; GARCH beats EWMA out of sample where volatility mean-reverts
- ▣ VaR 1% =  $-(\mu + \sigma_{t+1} q_{0.01}(z))$ : the volatility model and the quantile both matter

## Key Formulas

| Quantity                | Formula                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| ARCH( $q$ )             | $\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2, \quad \bar{\sigma}^2 = \omega / (1 - \sum \alpha_i)$          |
| GARCH(1,1)              | $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$ |
| Half-life               | $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$                                                                                       |
| GJR-GARCH               | $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$                                    |
| EGARCH                  | $\ln \sigma_t^2 = \omega + \alpha( z_{t-1}  - E z_{t-1} ) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$                          |
| Log-likelihood (Normal) | $\ell = -\frac{1}{2} \sum_t [\ln 2\pi + \ln \sigma_t^2 + (r_t - \mu)^2 / \sigma_t^2]$                                           |
| Forecast                | $E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2)$                               |
| QLIKE                   | $L = r_t^2 / h_t + \ln h_t$                                                                                                     |
| VaR 1%                  | $-(\mu + \sigma_{t+1} q_{0.01}(z))$                                                                                             |

## Check Yourself

- **Question:**  $\omega = 0.05$ ,  $\alpha = 0.05$ ,  $\beta = 0.90$ . What are the long-run variance and the half-life?
  - ▶ **Answer:**  $0.05/0.05 = 1$ ;  $\ln 0.5 / \ln 0.95 = 13.5$  days
- **Question:** GJR gives  $\hat{\alpha} = 0$ ,  $\hat{\gamma} = 0.2$ . What happens after a rise of 3%?
  - ▶ **Answer:** nothing beyond the decay  $\beta\sigma_t^2$ : only negative shocks raise the variance
- **Question:**  $Q(10)$  of  $\hat{z}_t^2$  has  $p = 0.40$ . Is the model correct?
  - ▶ **Answer:** we only know that no ARCH effect is left; asymmetry or the wrong distribution may remain
- Next: [Chapter 10](#), VaR, ES and backtesting

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